

response of 40-500Hz and enhanced current measurement using an external clamp sensor. The multimeter is priced at ₹ 4999.

Kyoritsu KEW India Instruments
www.kew-ltd.co.in

Vector network analyser

The new ZNLE vector network analyser (VNA) from Rohde & Schwarz can be used for RF measurements to characterise components such as antennae, attenuators, filters and PCBs. This standalone instrument



weighs only 6kg and has a footprint of just 408x235mm², saving up to two thirds of space on the workbench compared with competitive VNAs. The two-port vector network analyser provides quick and accurate bidirectional measurements of S11, S21, S12 and S22 parameters. An optional GPIB interface is available for remote control of the analyser.

The instrument comes in two models with frequency ranges from 1MHz to 3GHz (model ZNLE3) and up to 6GHz (model ZNLE6).

Rohde & Schwarz
www.rohde-schwarz.com

Digital multimeter

MECO has introduced a 6000-count true-RMS pocket-size digital multimeter, model 108B + TRMS. This auto-ranging digital multimeter with LCD backlight can measure voltages up to 1000V DC and 750V AC (true-RMS) and current up to 10A AC (true-RMS) and DC. Basic accuracy for DC voltage is $\pm 1.0\%$ rdg + 10dgt, for AC voltage $\pm 1.0\%$ rdg + 3dgt, for DC current



$\pm 1.5\%$ rdg + 3dgt and for AC current $\pm 1.5\%$ rdg + 3dgt. It can perform resistance, capacitance, frequency, duty cycle and temperature measurement and diode and audible continuity tests.

Meco Instruments
www.mecoinst.com

MISCELLANEOUS

MIMO application framework update

National Instruments (NI) has announced multiple-antenna user equipment support for its LabVIEW communications MIMO application framework. This makes MIMO application framework the world's only commercially available physical layer reference design that powers massive MIMO prototyping that goes beyond just desktop simulations to fully functional 5G deployments.

Wireless researchers can pair the MIMO application framework with NI software-defined radio hardware to conduct real-time, over-the-air experiments for a wide range of MIMO research topics including multi-user MIMO, single-user MIMO and massive MIMO. This multi-FPGA physical layer reference design ships with LabVIEW Communications source code, which is fully reconfigurable and modifiable, making it possible to create a complete network of multiple antenna devices with minimal system integration or design effort.

National Instruments (NI)
www.ni.com

Programmable DC power supplies

Scientific has introduced a new series of 800W programmable DC power supplies in compact 1U size, with 0-600V and 0-80A models. These power supplies accept universal AC mains input of 90V-270V, and

feature PFC to ensure low harmonics and built-in protection against over-voltage, under-voltage, overload and over-temperature. Master slave



option allows the units to connect in series or parallel for higher power. The multi-range output is an added advantage over traditional fixed-output or fixed auto-range output characteristics. Twenty user-defined sequences can be saved internally, allowing independent automation without external programming. The units are available with analogue programming, USB, LAN, RS-232 and RS-485 options.

Scientific Mes Technik
www.scientificindia.com

Access control

Matrix has introduced a Bluetooth-based user identification technology. This access control technology



allows users to open doors by simply pairing their mobile's Bluetooth with the inherent Bluetooth of the door controller. Users have to pair their device only once, after which they can gain access by merely twisting their mobile phones in the vicinity of the entry/exit point.

The key features of Bluetooth-based access control include Mobile as user credential, contactless identification, processing of multiple user simultaneously, no RF cards required and encrypted communication.

Matrix Comsec
www.MatrixAccessControl.com